1. Trigonometry

1.1. Ratios

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\text{cosec}\theta = \frac{1}{\sin\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

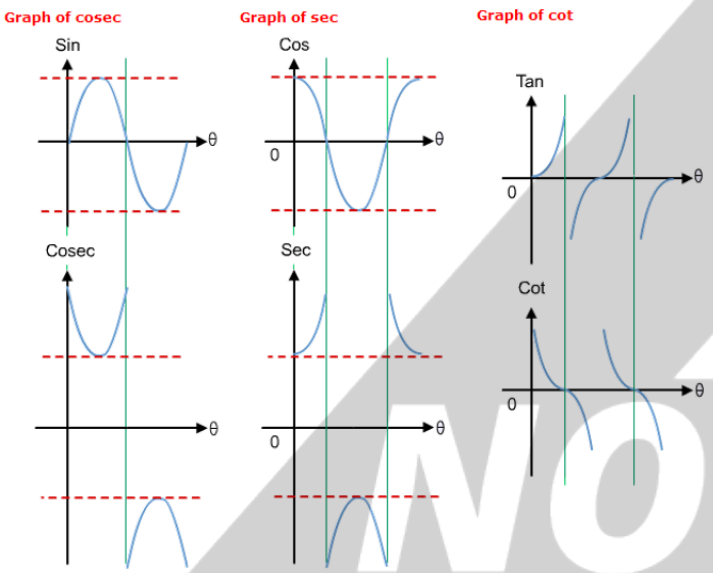
1.2. Identities

$$(\cos\theta)^2 + (\sin\theta)^2 \equiv 1$$

$$1 + (\tan\theta)^2 \equiv (\sec\theta)^2$$

$$(\cot\theta)^2 + 1 \equiv (\text{cosec}\theta)^2$$

1.3. Graphs



1.4. Double Angle Identities

$$\sin 2A \equiv 2\sin A\cos A$$

$$\cos 2A \equiv (\cos A)^2 - (\sin A)^2$$

$$\equiv 2(\cos A)^2 - 1 \equiv 1 - 2(\sin A)^2$$

$$\tan 2A \equiv \frac{2\tan A}{1-(\tan A)^2}$$

1.5. Addition Identities

$$\sin(A \pm B) \equiv \sin A\cos B \pm \cos A\sin B$$

$$\cos(A \pm B) \equiv \cos A\cos B \mp \sin A\sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

1.6. Changing Forms

$a\sin x \pm b\cos x \Longleftrightarrow R\sin(x \pm \alpha)$

$a\cos x \pm b\sin x \Longleftrightarrow R\cos(x \mp \alpha)$

Where $R = \sqrt{a^2+b^2}$ and

$R\cos\alpha = a \mid R\sin\alpha = b$ with $0 < \alpha < \frac{1}{2}\pi$

{S13-P33} Question 9:

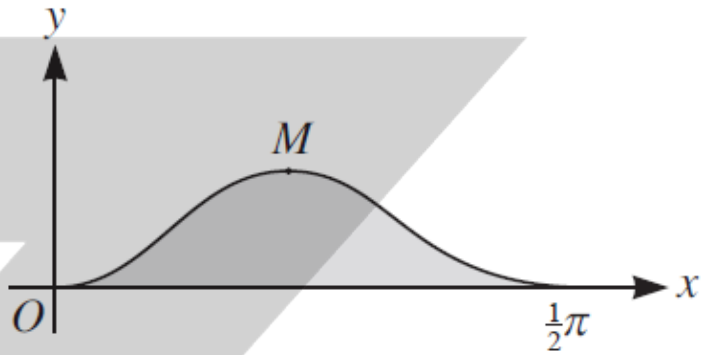


Diagram shows curve, $y = \sin^2 2x\cos x$, for $0 \le x \ge \frac{\pi}{2}$, and M is maximum point. Find the x coordinate of M.

Solution:

Use product rule to differentiate:

$$u = \sin^2 2x \qquad v = \cos x$$

$$u' = 4\sin 2x\cos 2x \qquad v' = -\sin x$$

$$\frac{dy}{dx} = u'v + uv'$$

$$\frac{dy}{dx} = (4\sin 2x\cos 2x)(\cos x) + (\sin^2 2x)(-\sin x)$$

$$\frac{dy}{dx} = 4\sin 2x\cos 2x\cos x - \sin^2 2x\sin x$$

Use following identities:

$$\cos 2x = 2\cos^2 x - 1$$

$$\sin 2x = 2\sin x\cos x$$

$$\sin^2 x = 1 - \cos^2 x$$

Equating to 0*:*

$$\frac{dy}{dx} = 0$$

$$\therefore 4\sin 2x\cos 2x\cos x - \sin^2 2x\sin x = 0$$

$$4\sin 2x\cos 2x\cos x = \sin^2 2x\sin x$$

Cancel $\sin 2x$ *on both sides*

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$$4\cos 2x\cos x = \sin 2x\sin x$$

Substitute identities

$$4\left(2\cos^2 x - 1\right)\cos x = (2\sin x\cos x)\sin x$$

Cancel $\cos x$ *and constant* 2 *from both sides*

$$4\cos^2 x - 2 = \sin^2 x$$

Use identity

$$4\cos^2 x - 2 = 1 - \cos^2 x$$

$$5\cos^2 x = 3$$

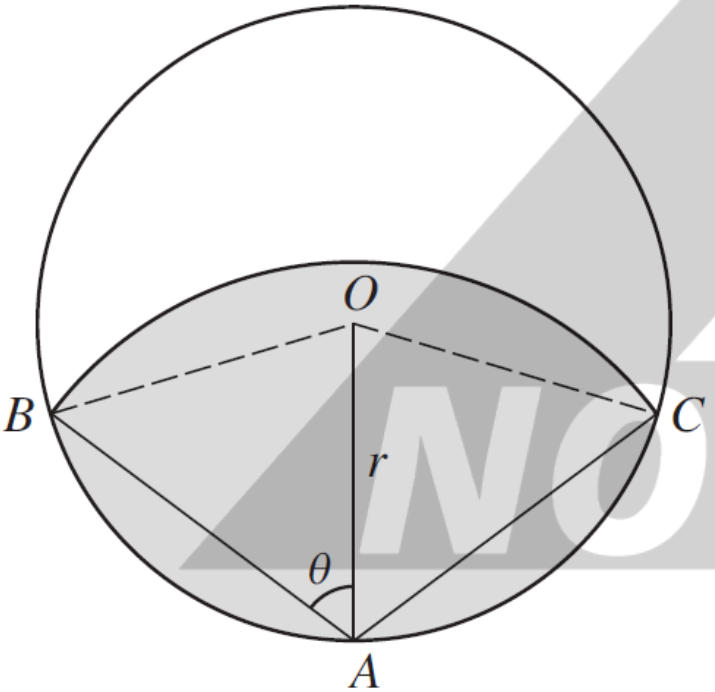
$$\cos^2 x = \frac{3}{5}$$

$$\cos x = 0.7746$$

$$x = \cos^{-1}(0.7746)$$

$$x = 0.6847 \approx 0.685$$

{W13-P31} Question 6:



A is a point on circumference of a circle center O, radius r. A circular arc, center A meets circumference at B & C. Angle OAB is θ radians. The area of the shaded region is equal to half the area of the circle.

Show that:

$$\cos 2\theta = \frac{2\sin 2\theta - r}{4\theta}$$

Solution:

First express area of sector OBAC

$$Sector\ Area = \frac{1}{2}\theta r^2$$

$$OBAC = \frac{1}{2}(2\pi - 4\theta)r^2 = (\pi - 2\theta)r^2$$

Now express area of sector ABC

$$ABC = \frac{1}{2}(2\theta)(\text{Length of BA})^2$$

Express BA *using sine rule*

$$BA = \frac{r\sin(\pi - 2\theta)}{\sin\theta}$$

Use double angle rules to simplify this expression

$$BA = \frac{r\sin 2\theta}{\sin\theta}$$

$$= \frac{2r\sin\theta\cos\theta}{\sin\theta}$$

$$= 2r\cos\theta$$

Substitute back into initial equation

$$ABC = \frac{1}{2}(2\theta)(2r\cos\theta)^2$$

$$ABC = 4\theta r^2\cos^2\theta$$

Now express area of kite ABOC

$$ABOC = 2 \times Area\ of\ Triangle$$

$$ABOC = 2 \times \frac{1}{2}r^2\sin(\pi - 2\theta)$$

$$= r^2\sin(\pi - 2\theta)$$

Finally, the expression of shaded region equated to half of circle

$$4r^2\theta\cos^2\theta + r^2(\pi - 2\theta) - r^2\sin(\pi - 2\theta) = \frac{1}{2}\pi r^2$$

Cancel our r^2 *on both sides for all terms*

$$4\theta\cos^2\theta + \pi - 2\theta - (\sin\pi\cos 2\theta + \sin 2\theta\cos\pi) = \frac{1}{2}\pi$$

Some things in the double angle cancel out

$$4\theta\cos^2\theta + \pi - 2\theta - \sin 2\theta = \frac{1}{2}\pi$$

Use identity here

$$4\theta\left(\frac{\cos 2\theta + 1}{2}\right) + \pi - \sin 2\theta - 2\theta = \frac{1}{2}\pi$$

$$4\theta\cos 2\theta + 4\theta + 2\pi - 2\sin 2\theta - 4\theta = \pi$$

Clean up

$$4\theta\cos 2\theta + 2\pi - 2\sin 2\theta = \pi$$

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$$4\theta \cos 2\theta = 2\sin 2\theta - \pi$$

$$\cos 2\theta = \frac{2\sin 2\theta - \pi}{4\theta}$$

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